```
$ sntp -d time.euro.apple.com
...
sntp_exchange {
        result: 0 (Success)
        header: 24 (li:0 vn:4 mode:4)
       stratum: 01 (1)
          poll: 06 (64)
     precision: FFFFFFEB (4.768372e-07)
...
```

You should see multiple sets of debug output like the above. There's no need to go too deeply into how all this works; at the moment, it's enough for you to know that these tools and services exist and there's plenty of information and support for NTP out there on the web when you're ready to know more, or in the sticky predicament of *needing* to know more!

The tz Database

Another problematic aspect to timekeeping in computing is knowing *where* you are. You might have servers located in different geographical regions; they might even be nicely synchronized to UTC, but ask them what the time is and you'll get two different answers, because of course they are in different time zones. Naturally, there are systems and standards to manage time zones.

The `tz database` (a.k.a. `tzdata`, `zoneinfo`, or the `IANA time zone db`, occasionally known as the `Olson database` after its founder Arthur Olson) is fairly ubiquitous and is the basis for how PHP parses time zone information. Time zones in the `tz database` are canonically identified with strings such as `Europe/London` or `America/New York`, or their abbreviated forms such as `PST` and `GMT`. Note that the canonical regional names will aggregate to abbreviated names. For example, the regions `Europe/Paris` and `Europe/Amsterdam` are both in the `CET` time zone. More comprehensive information can be found at the following locations:

- https://www.php.net/manual/en/timezones.php
- https://en.wikipedia.org/wiki/Tz_database
- https://en.wikipedia.org/wiki/List_of_time_zone_abbreviations

Right then, it's time to get our hands dirty with… ahem… time.